

George Paul

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Statistician and machine learning researcher (MSc Data Science, Distinction) experienced in Bayesian methodology, uncertainty quantification and modern AI systems. My research path runs from an MSc dissertation that grew into a first-author methodological paper (prepared for submission to the *Electronic Journal of Statistics*), through an applied research role in industry where I moved into fine-tuning and evaluating vision-language and language models; work that became a peer-reviewed EC3 2026 paper. That transition gave me hands-on experience of the gap between how models are evaluated on small held-out sets and how they behave in deployment.

PUBLICATIONS & PAPERS UNDER REVIEW

- **Paul, G.** and Tipping, M. E. (2026). The Adaptive Elastic Horseshoe Prior: Data-Driven Regularisation for Mixed Sparse–Dense Feature Structures via Momentum-Based Empirical Bayes. *Manuscript prepared for submission, Electronic Journal of Statistics*.
- [Co-authors incl. **Paul, G.**] (2026). Benchmarking Deep Learning Approaches for AEC Engineering Drawing Layout Detection and Information Extraction. *Proceedings of the European Conference on Computing in Construction (EC3), Corfu, Greece, 2026*. Peer-reviewed; contribution: vision-language model (Florence-2) fine-tuning methodology and evaluation.

RESEARCH EXPERIENCE

The Adaptive Elastic Horseshoe (AEH) Prior Feb. 2025 – Present
First-author journal paper prepared for submission to EJS, with Prof. Michael Tipping University of Bath

- **Methodology:** Introduced a global–local shrinkage prior that learns the balance between sparsity-inducing (ℓ_1) and grouping (ℓ_2) regularisation from data, with hyperparameters updated via momentum-driven empirical Bayes during HMC warm-up.
- **Theory:** Established the adaptive updates are approximate M-steps of an EM algorithm on a mean-field variational lower bound, and established minimax-optimal posterior contraction for the sparse normal-means problem, matching the horseshoe rate of van der Pas et al. (2014).
- **Experimental design & evaluation:** Designed eight simulation studies, including decision-boundary sweeps at $p=20$ and $p=200$ identifying precisely when AEH outperforms the horseshoe (within-group correlation $\rho \gtrsim 0.6$, group size ≥ 5), plus a real-data application to the Efron et al. (2004) diabetes benchmark.
- Originated as the MSc dissertation (Feb. 2025); continued independently alongside full-time work after submission to develop it into a first-author manuscript, strengthening the theoretical results and evaluation ahead of submission.

Benchmarking Deep Learning Approaches for AEC Engineering Drawing Layout Detection 2025 – 2026
Peer-reviewed conference paper, EC3 2026 — contribution: VLM fine-tuning methodology & evaluation Buro Happold

- **Representation learning:** Delivered the vision-language model arm of a five-architecture benchmark (Faster R-CNN, YOLOv12, RF-DETR, Florence-2, DocLayout-YOLO) for AEC drawing layout detection, fine-tuning Florence-2 via parameter-efficient LoRA adaptation on a custom 551-drawing dataset (FELD).
- **Principled evaluation & negative results:** Designed the controlled, held-out protocol comparing all five architectures under matched conditions. Florence-2’s multi-modal pretraining transferred poorly to AEC drawings’ line-based logic ($mAP_{50} = 0.261$ vs. 0.949 for the best detector), contributing to a broader *domain-interference* finding: weights pretrained on general-document layouts underperformed models without that prior, showing representations learned on one distribution can actively harm transfer to a structurally different one.

PROFESSIONAL EXPERIENCE

Buro Happold Sep. 2025 – Present
Data Science Analyst London, UK

- **LLM/VLM fine-tuning & evaluation:** Fine-tuned vision-language and language models (PyTorch, LoRA, SWIFT; Azure ML and Hugging Face model APIs) to extract structured information from 1,500+ technical documents; owned the labelling framework and controlled evaluation protocol (extraction F1 > 0.84). Work contributed to the peer-reviewed EC3 2026 paper above.
- **Anomaly & risk detection at scale:** Built a geospatial risk-scoring system combining five years of time-series outage data with satellite-derived features (40+ engineered spatio-temporal variables), producing failure-risk rankings across 10,000+ assets used in capital-allocation decisions. Explicitly tuned the precision/recall operating point (87% precision, 82% recall out-of-sample) against the asymmetric cost of false positives versus missed failures.
- **Validation infrastructure:** Developed end-to-end model-validation pipelines in Python on Azure (cross-validation, hypothesis testing, sensitivity analysis and scenario evaluation) with explicit checks for robustness to distribution shift.

Green Place Assets Jun. 2024 – Jul. 2025
Data Science Intern Remote

- Designed a building-performance index combining time-series volatility, peer benchmarking and multivariate scoring to rank underperformance signals; built gradient-boosting and random-forest models on tabular and geospatial data (out-of-sample $R^2 = 0.81$), validated via cross-validation.

EDUCATION

University of Bath

Sep. 2024 – Aug. 2025

MSc Data Science — Distinction (overall 80.7%)

Bath, UK

- Core modules: Statistics (94%), Machine Learning (91%), Bayesian Machine Learning (88%), Research (81%). Training in statistical inference, probabilistic modelling, time-series analysis, deep learning and reinforcement learning.
- Dissertation (supervised by Prof. Michael Tipping): developed a novel Bayesian shrinkage prior with full HMC inference; extended post-degree into a first-author journal paper now prepared for submission to the *Electronic Journal of Statistics*.

University of Manchester

Sep. 2021 – Jun. 2024

BSc Mathematics — 2:1

Manchester, UK

- Probability theory, mathematical statistics, stochastic processes, numerical methods, linear algebra, graph theory; scientific computing in Python and MATLAB.

TECHNICAL SKILLS

- **Statistical machine learning & Uncertainty Quantification:** Bayesian inference, variational inference, empirical Bayes, shrinkage priors, Gaussian processes, hypothesis testing, uncertainty quantification and calibration (PICP, posterior predictive checks), Monte Carlo methods, time-series analysis.
- **Machine learning:** Transformer-based models and LLM/VLM fine-tuning, representation learning, supervised/unsupervised learning, ensemble methods, anomaly detection and classification under asymmetric error costs, cross-validation and experimental design, SHAP-based interpretation.
- **Programming:** Python (NumPy, SciPy, scikit-learn, statsmodels, PyTorch, JAX, Hugging Face), REST/model API integration, SQL, MATLAB, LaTeX, Git/GitHub, Azure ML.